Aerofit Customer Profiling Analysis

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1. Defining Problem Statement and Analysing Basic Metrics

Problem Statement

AeroFit wants to understand the characteristics of customers purchasing KP281, KP481, and KP781 treadmills and identify factors associated with product purchase to provide better recommendation of the treadmills to the new customers.

```
# Import Libraries and Dataset

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

sns.set_theme(style="whitegrid")

url =
"https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/125/original/aerofit_treadmill.csv?1639992749"

df = pd.read_csv(url)

display(df.head())
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
# Shape and Structure

print("Shape of the dataset:", df.shape)

df.info()

# Data Types

print("\nData Types:")

print(df.dtypes)

# Convert categorical variables:

categorical_cols = ['Product', 'Gender', 'MaritalStatus']

df[categorical_cols] = df[categorical_cols].astype('category')

print("\nData Types After Conversion:")

print(df.dtypes)

# Statistical Summary

display(df.describe().T)
```

```
Shape of the dataset: (180, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product        180 non-null    object
 1   Age            180 non-null    int64
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
Data Types:
Product          object
Age               int64
Gender           object
Education         int64
MaritalStatus    object
Usage             int64
Fitness           int64
Income            int64
Miles             int64
dtype: object

Data Types After Conversion:
Product          category
Age                 int64
Gender           category
Education           int64
MaritalStatus    category
Usage               int64
Fitness             int64
Income              int64
Miles               int64
dtype: object
```

	count	mean	std	min	25%	50%	75%	max
Age	180.0	28.788889	6.943498	18.0	24.00	26.0	33.00	50.0
Education	180.0	15.572222	1.617055	12.0	14.00	16.0	16.00	21.0
Usage	180.0	3.455556	1.084797	2.0	3.00	3.0	4.00	7.0
Fitness	180.0	3.311111	0.958869	1.0	3.00	3.0	4.00	5.0
Income	180.0	53719.577778	16506.684226	29562.0	44058.75	50596.5	58668.00	104581.0
Miles	180.0	103.194444	51.863605	21.0	66.00	94.0	114.75	360.0

Insights

- The dataset contains 180 rows and 9 columns.
- Product, Gender, and MaritalStatus are categorical variables.
- Age ranges from 18 to 50 years.
- Income ranges from $29,562 to $104,581.
- Usage ranges from 2 to 7 times per week.
- Fitness ranges from 1 to 5.
- Miles range from 21 to 360.

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2. Non-Graphical Analysis: Value Counts and Unique Attributes

We first check the unique values and frequency of the categorical/discrete variables.

```
for col in ['Product', 'Gender', 'MaritalStatus', 'Education', 'Usage',
'Fitness']:
    print(f"\n--- {col} ---")
    print("Number of unique values:", df[col].nunique())
    print(df[col].value_counts().sort_index())

print("\nProduct Value Counts:")
print(df['Product'].value_counts())

print("\nGender Value Counts:")
print(df['Gender'].value_counts())
```

```
print("\nMarital Status Value Counts:")
print(df['MaritalStatus'].value_counts())
```

```
--- Product ---
Number of unique values: 3
Product
KP281    80
KP481    60
KP781    40
Name: count, dtype: int64

--- Gender ---
Number of unique values: 2
Gender
Female     76
Male      104
Name: count, dtype: int64

--- MaritalStatus ---
Number of unique values: 2
MaritalStatus
Partnered    107
Single        73
Name: count, dtype: int64
```

```
--- Education ---
Number of unique values: 8
Education
12     3
13     5
14    55
15     5
16    85
18    23
20     1
21     3
Name: count, dtype: int64

--- Usage ---
Number of unique values: 6
Usage
2    33
3    69
4    52
5    17
6     7
7     2
Name: count, dtype: int64

--- Fitness ---
Number of unique values: 5
Fitness
1     2
2    26
3    97
4    24
5    31
Name: count, dtype: int64
```

```
Product Value Counts:
Product
KP281    80
KP481    60
KP781    40
Name: count, dtype: int64

Gender Value Counts:
Gender
Male      104
Female     76
Name: count, dtype: int64

Marital Status Value Counts:
MaritalStatus
Partnered    107
Single        73
Name: count, dtype: int64
```


```
Product Value Counts:
Product
KP281    80
KP481    60
KP781    40
Name: count, dtype: int64

Gender Value Counts:
Gender
Male      104
Female     76
Name: count, dtype: int64

Marital Status Value Counts:
MaritalStatus
Partnered    107
Single        73
Name: count, dtype: int64
```


Insights

- KP281: 80 customers
- KP481: 60 customers
- KP781: 40 customers
- Male: 104 customers
- Female: 76 customers

- Partnered: 107 customers
- Single: 73 customers
- Usage of 3 times/week is the most common with 69 customers.
- Fitness level 3 is the most common with 97 customers.

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3. Visual Analysis — Univariate & Bivariate

This section contains only the visual analysis required by the evaluation criteria and the business questions.

3.1 Univariate Analysis — Categorical Variables

```
# Summary table with counts and percentages
summary_table = pd.DataFrame({
    'Attribute': ['Product', 'Product', 'Product','Gender', 'Gender',
        'Marital Status', 'Marital Status'],
    'Category': ['KP281', 'KP481', 'KP781','Male', 'Female',
'Partnered', 'Single'],
    'Count': [(df['Product'] == 'KP281').sum(),
        (df['Product'] == 'KP481').sum(),
        (df['Product'] == 'KP781').sum(),
        (df['Gender'] == 'Male').sum(),
        (df['Gender'] == 'Female').sum(),
        (df['MaritalStatus'] == 'Partnered').sum(),
        (df['MaritalStatus'] == 'Single').sum()
    ]
})

summary_table['Percentage'] = (
    summary_table['Count'] / len(df) * 100
).round(2)

display(summary_table)

#Bar Graphs
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
```

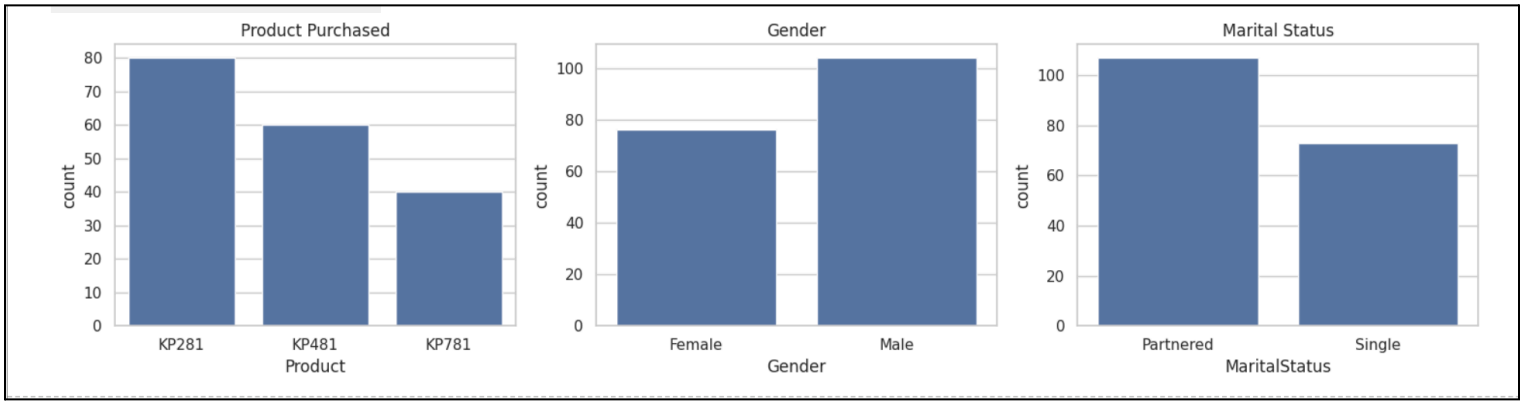
```
sns.countplot(data=df, x='Product', ax=axes[0])
axes[0].set_title('Product Purchased')

sns.countplot(data=df, x='Gender', ax=axes[1])
axes[1].set_title('Gender')

sns.countplot(data=df, x='MaritalStatus', ax=axes[2])
axes[2].set_title('Marital Status')

plt.tight_layout()
plt.show()
```

	Attribute	Category	Count	Percentage
0	Product	KP281	80	44.44
1	Product	KP481	60	33.33
2	Product	KP781	40	22.22
3	Gender	Male	104	57.78
4	Gender	Female	76	42.22
5	Marital Status	Partnered	107	59.44
6	Marital Status	Single	73	40.56



Insights

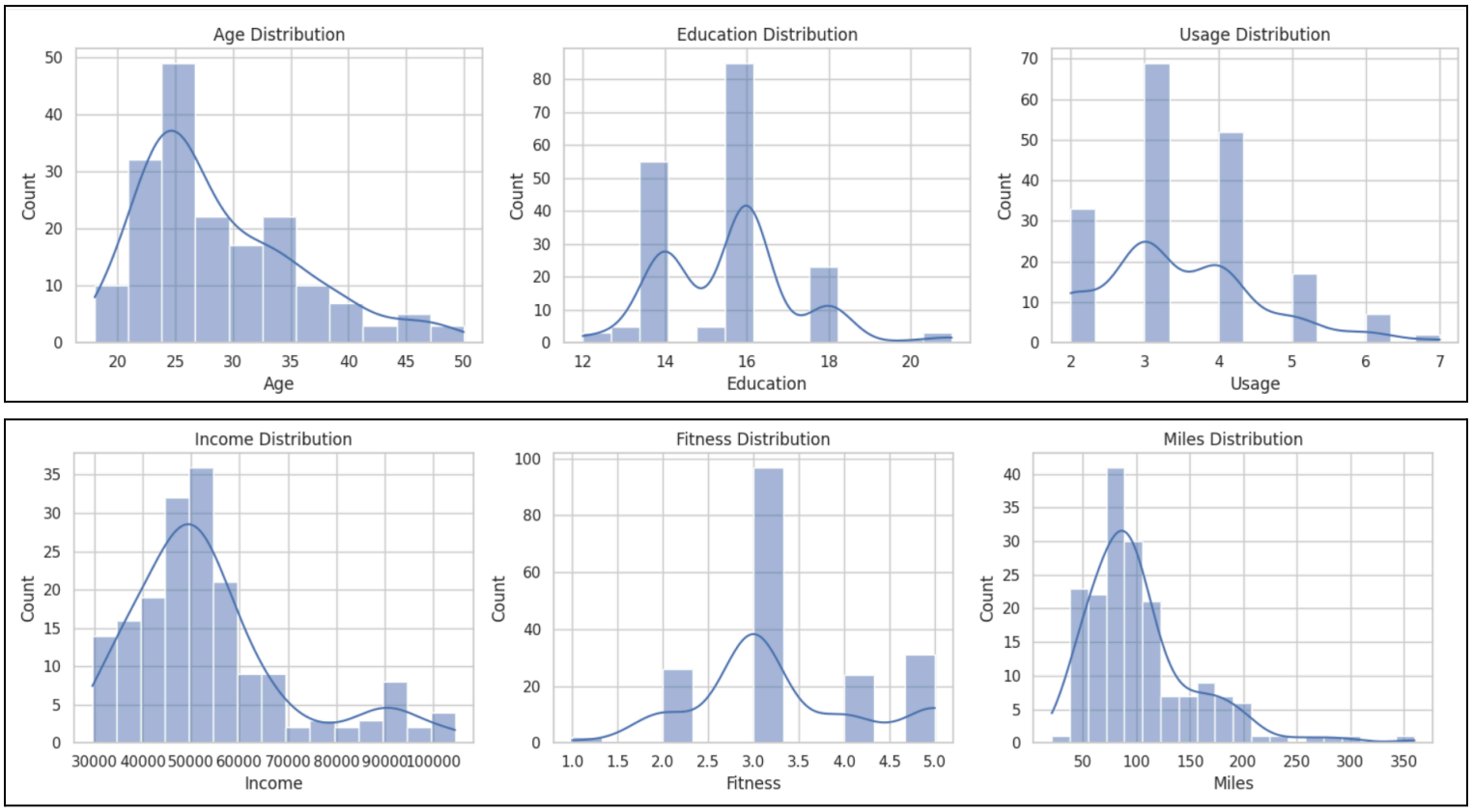
- **KP281 has the highest customer share**, with **80 customers (44.44%)**, making it the most commonly purchased product.

- **Male customers represent 57.78%** of the dataset, indicating a slightly male-dominated customer base.
- **Partnered customers represent 59.44%** of the dataset, while single customers account for 40.56%, indicating a slightly higher representation of partnered customers.
- Overall, **KP281 has the broadest customer base**, while the dataset is slightly more concentrated toward **male and partnered customers**.

3.2 Univariate Analysis — Continuous Variables

```
continuous_cols = ['Age','Education','Usage','Income','Fitness',
    'Miles']

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
for ax, col in zip(axes.flatten(), continuous_cols):
    sns.histplot(data=df, x=col, kde=True, ax=ax)
    ax.set_title(f'{col} Distribution')
plt.tight_layout()
plt.show()
```

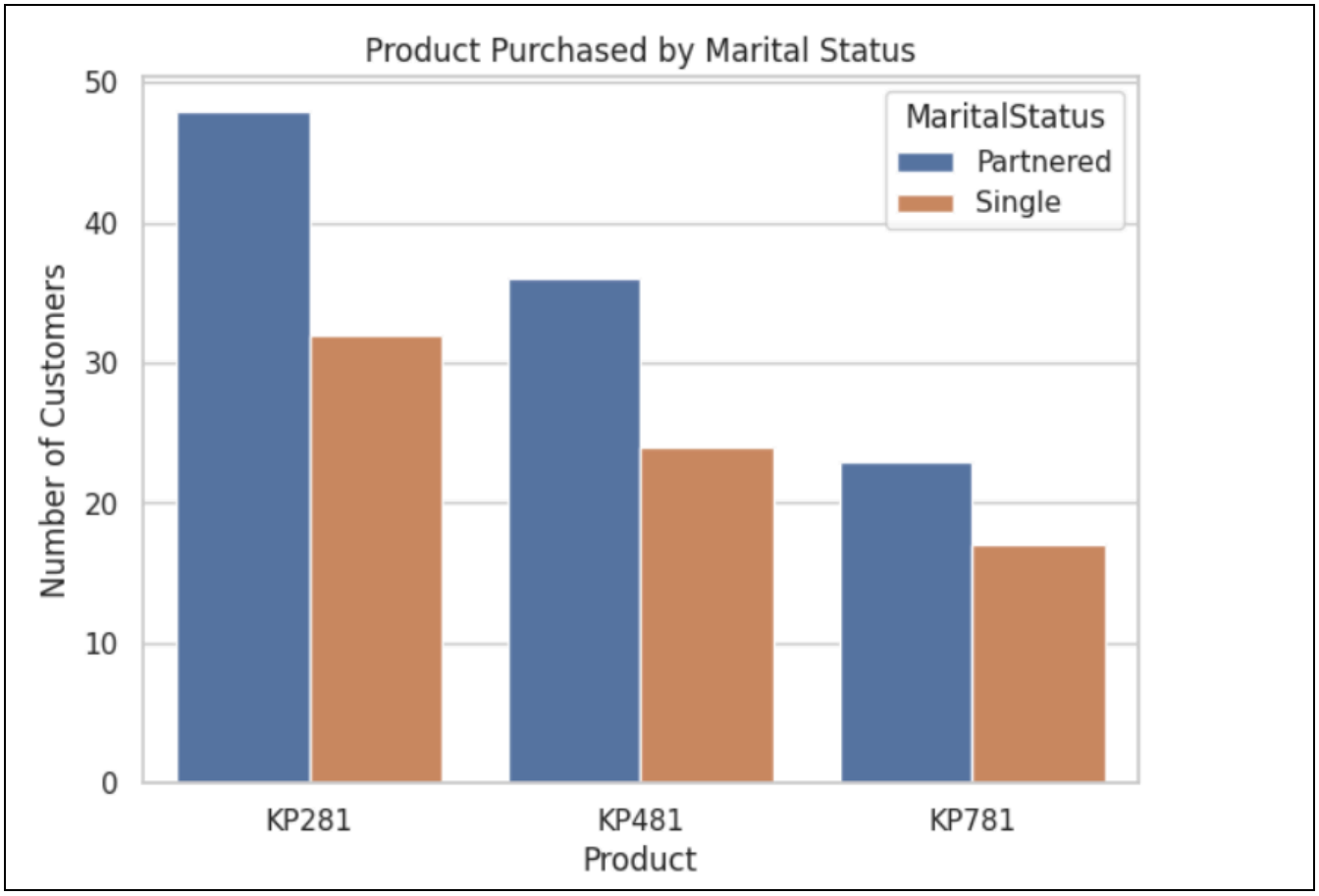


Insights

- **Age:** Most customers are in their **20s and early 30s**, indicating that the customer base is relatively young.
- **Education:** Education is concentrated around **14–16 years**, with **16 years** being the most common level.
- **Usage:** Customer usage is concentrated around **2–4 times per week**, with **3 times per week** being the most common.
- **Income:** Income is **right-skewed**, with most customers concentrated at lower income levels and a smaller number of customers having **substantially higher incomes**.
- **Fitness: Fitness level 3** is the most common, indicating that the majority of customers fall into the **moderate-fitness category**.
- **Miles:** Miles are also **right-skewed**, with most customers recording relatively lower mileage and a smaller group showing **substantially higher expected usage**.

3.3 Marital Status vs Product

```
plt.figure(figsize=(7, 5))
sns.countplot(data=df,x='Product',hue='MaritalStatus')
plt.title('Product Purchased by Marital Status')
plt.xlabel('Product')
plt.ylabel('Number of Customers')
plt.show()
```



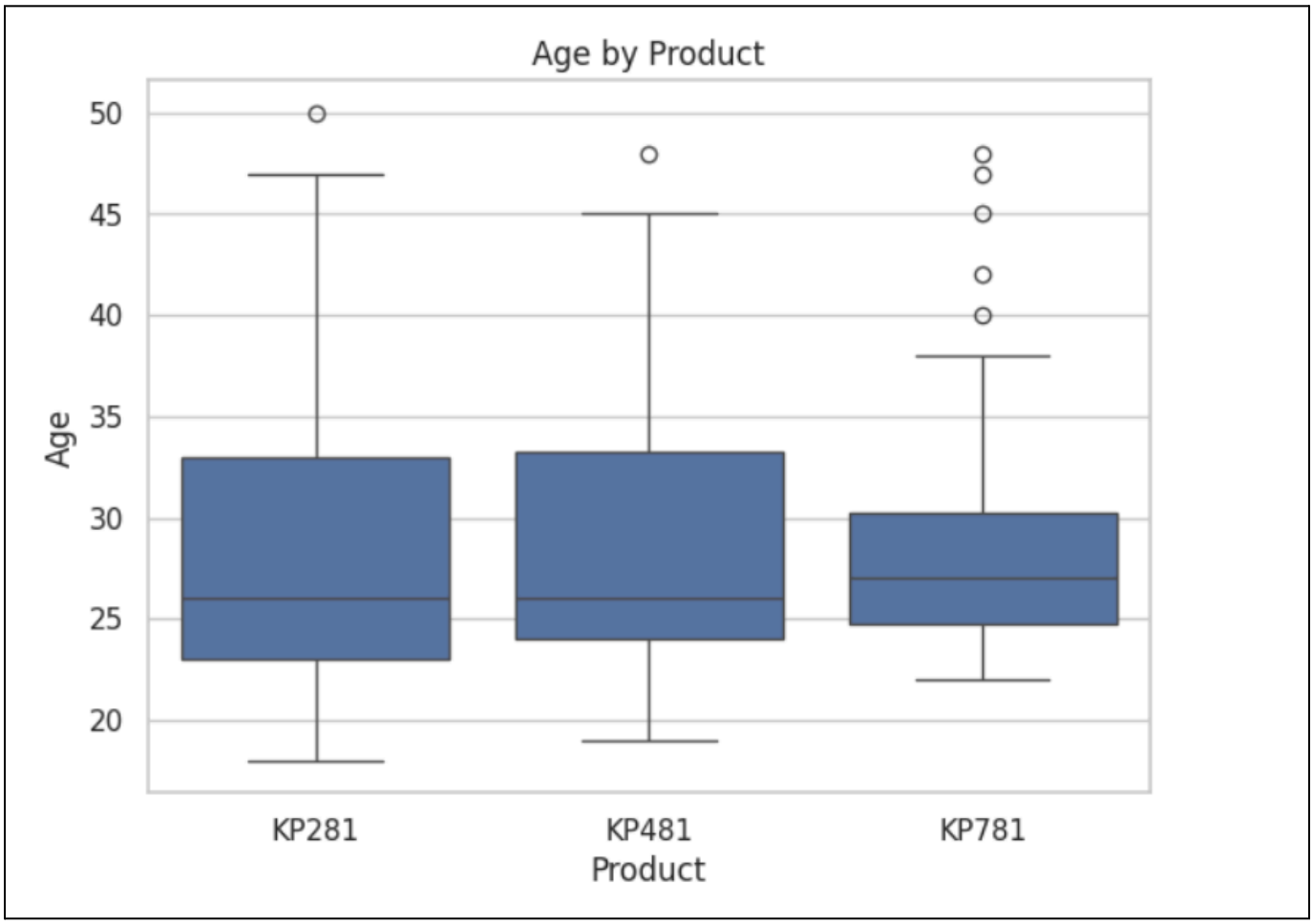
Insights

- **KP281:** 48 partnered and 32 single customers, with **60.00% partnered**.
- **KP481:** 36 partnered and 24 single customers, with **60.00% partnered**.
- **KP781:** 23 partnered and 17 single customers, with **57.50% partnered**.
- The partnered share varies only slightly across products, from **57.50% to 60.00%**, showing a **similar marital-status distribution across all three products**.
- Therefore, **marital status does not appear to be a strong factor in differentiating product choice**, unlike characteristics such as **income, usage, fitness, and miles**.

3.4 Age vs Product

```
plt.figure(figsize=(7, 5))
sns.boxplot(data=df,x='Product',y='Age')
plt.title('Age by Product')
plt.xlabel('Product')
plt.ylabel('Age')
```

```
plt.show()
```



Insights

- **Average age is very similar across all three products:** KP281 = **28.55 years**, KP481 = **28.90 years**, and KP781 = **29.10 years**.
- The difference between the youngest and oldest product averages is only **0.55 years**, indicating **very little variation in average age across product groups**.
- The boxplots also show **substantial overlap in age distributions** across KP281, KP481, and KP781.
- Therefore, **age does not appear to be a strong factor in differentiating product choice**. Other characteristics such as **income, usage, fitness, and miles** show clearer differences across products.

3.5 Other Numerical Variables vs Product

These boxplots help identify which customer characteristics differ across products.

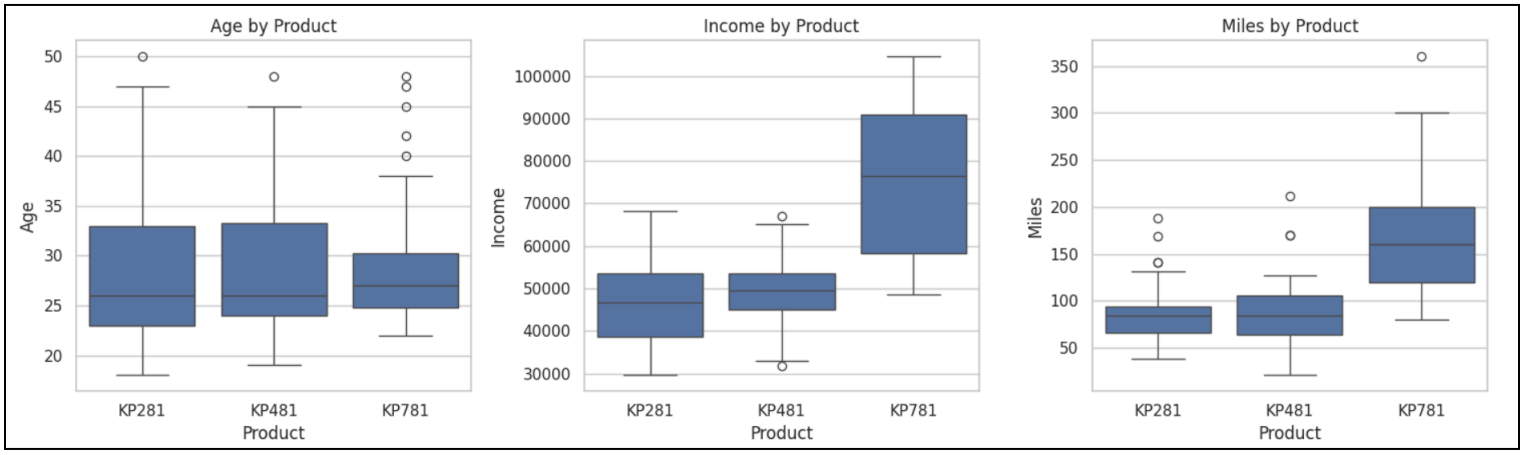
```
boxplot_vars = ["Age", "Income", "Miles", "Usage", "Fitness"]

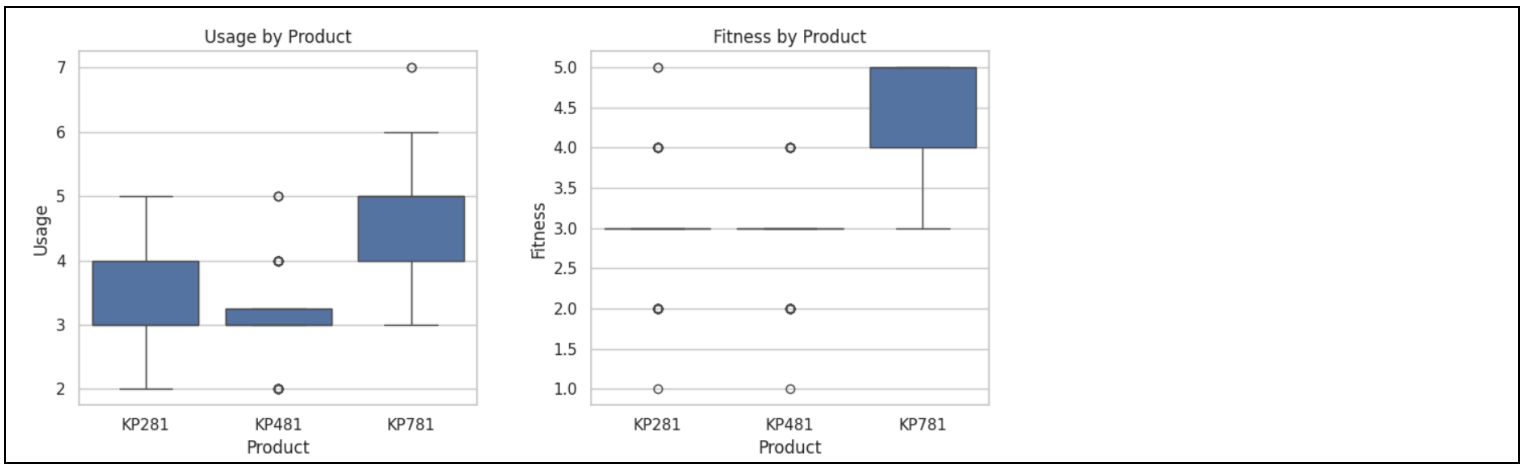
fig, axes = plt.subplots(2, 3, figsize=(15, 9))

for ax, col in zip(axes.flatten(), boxplot_vars):
    sns.boxplot(data=df, x="Product", y=col, ax=ax)
    ax.set_title(f"{col} by Product")

# Hide the unused sixth subplot
axes.flatten()[-1].set_visible(False)

plt.tight_layout()
plt.show()
```





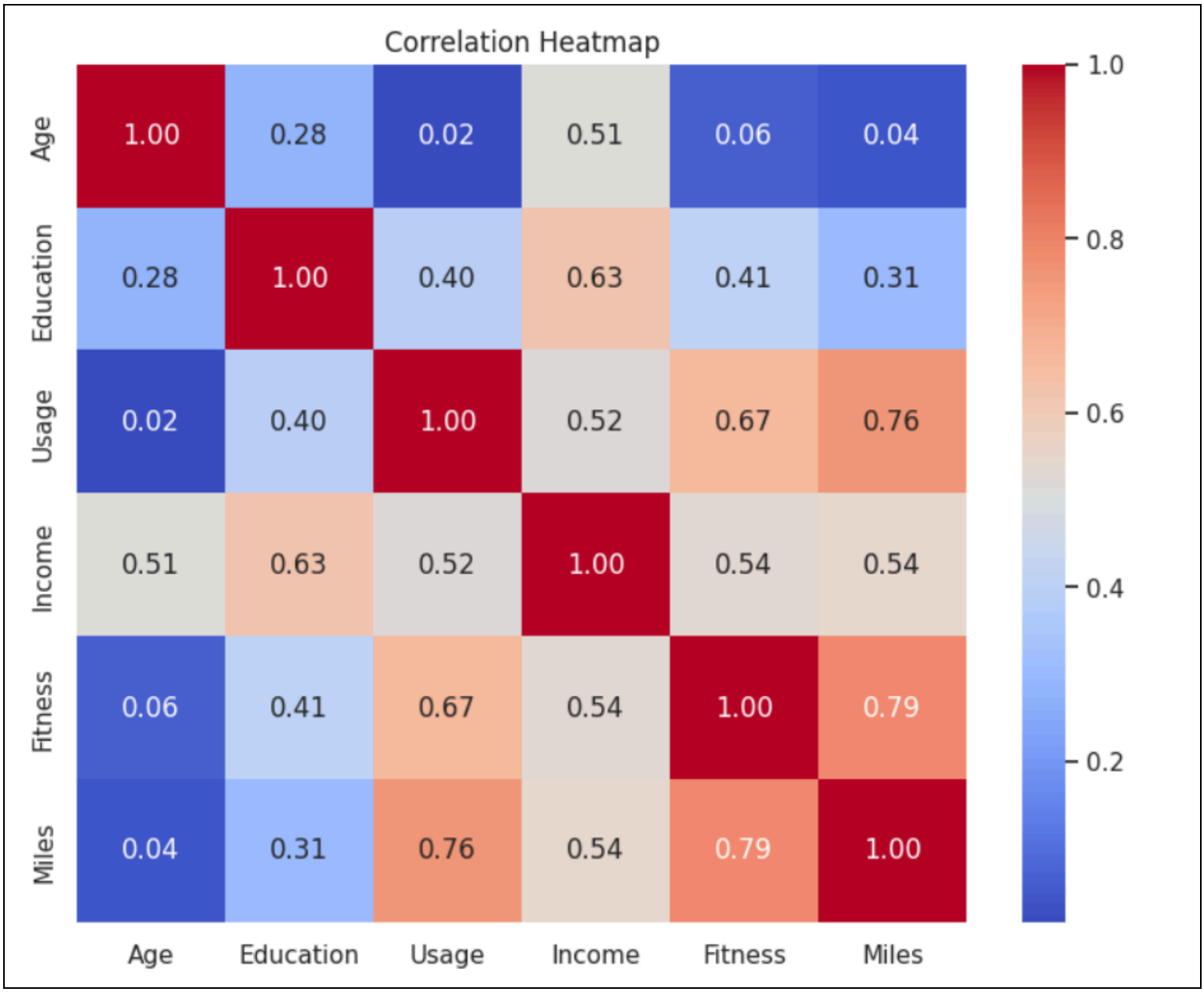
Insights

- **KP781 customers have substantially higher purchasing power**, with an average income of **$75,442**, compared with **$46,418 for KP281**. This suggests that KP781 is more closely associated with **higher-income customers**.
- **KP781 customers show higher treadmill usage**, averaging **4.78 times per week**, compared with about **3.1 times per week for KP281/KP481**. This indicates that KP781 is more commonly used by **frequent treadmill users**.
- **KP781 customers have a higher average fitness level of 4.63**, compared with about **3 for KP281/KP481**, suggesting that KP781 is more strongly associated with **fitness-oriented and more active customers**.
- **KP781 customers report substantially higher average mileage at 166.90 miles**, compared with **82.79 miles for KP281**. This further supports the pattern that KP781 attracts **customers with higher expected treadmill usage**.
- Overall, **Income, Usage, Fitness, and Miles show stronger differences across products than Age**, indicating that **purchasing power and activity level are more useful for differentiating product segments**.

3.6 Correlation Heatmap

```
numeric_cols = ['Age','Education','Usage','Income','Fitness',
    'Miles']

plt.figure(figsize=(9, 7))
sns.heatmap(df[numeric_cols].corr(),annot=True,fmt='.2f',
    cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



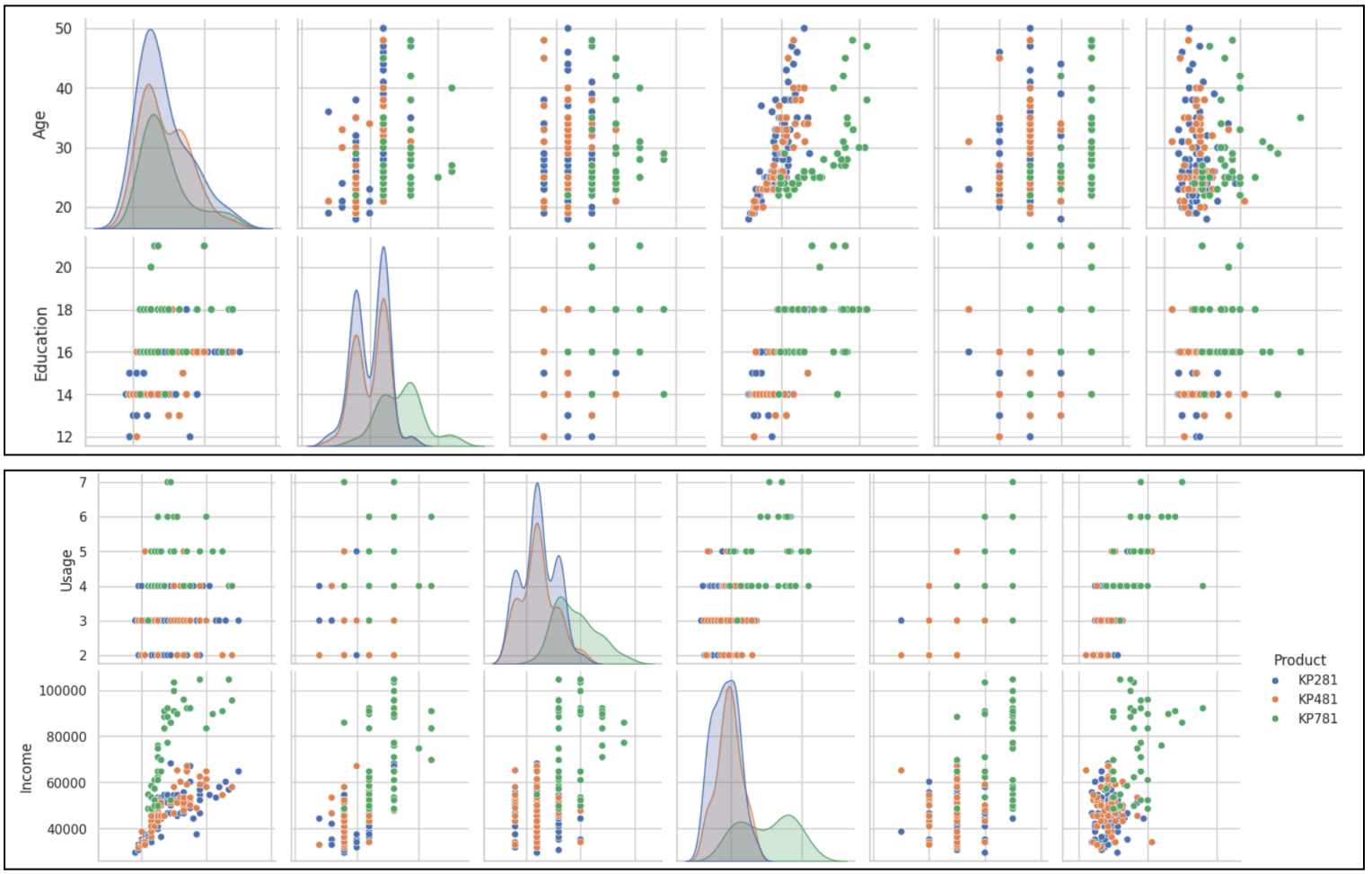
Insights

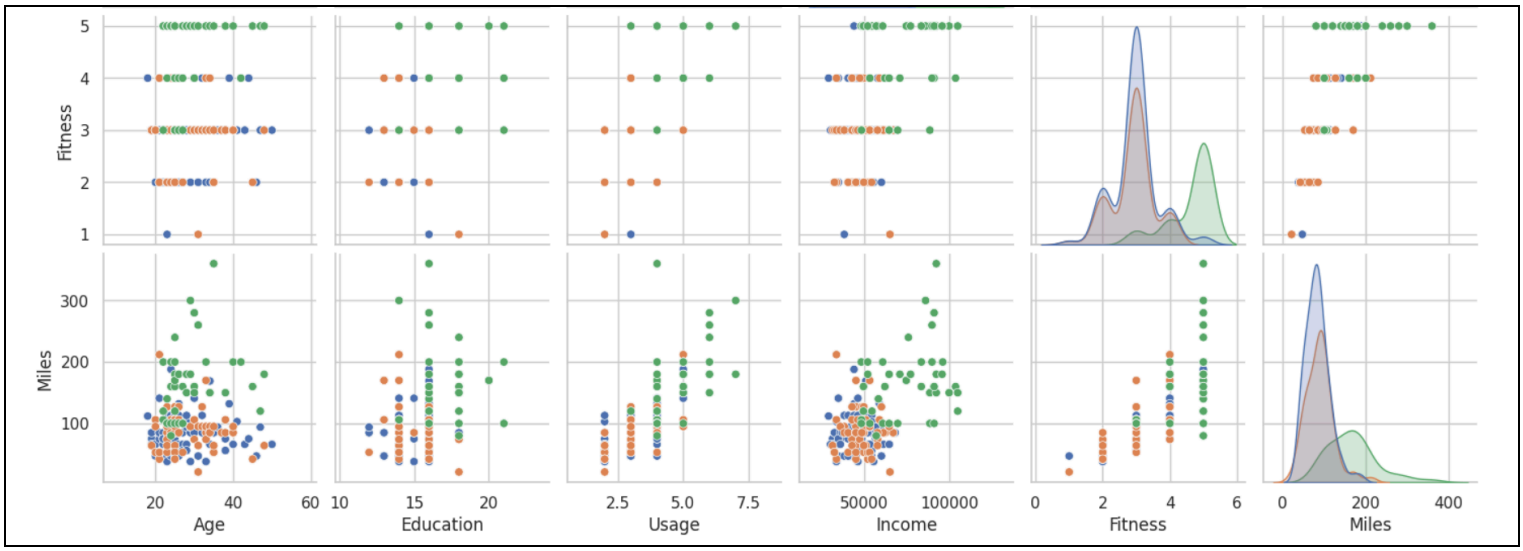
- **Usage and Miles have the strongest positive relationship (0.76**): customers who use the treadmill more frequently also tend to record higher expected mileage.
- Usage and Fitness have the second strongest relationship (0.67): customers who exercise more frequently tend to have higher fitness levels.
- Education and Income are positively related (0.63): customers with higher education levels generally have higher incomes.
- Income and Usage show a moderate positive relationship (0.52): higher-income customers tend to report more frequent treadmill usage.

- Age has almost no relationship with Miles (0.03): expected treadmill mileage is not meaningfully associated with customer age.

3.7 Pairplot

```
sns.pairplot(df,vars=['Age','Education','Usage','Income',
'Fitness','Miles'],hue='Product')
plt.show()
```





Insights

- **Age:** Age shows substantial overlap across KP281, KP481, and KP781 across the major numeric variables. Therefore, age does not appear to be a strong differentiator of product choice compared with income, usage, fitness, and miles.
- **Education and Income:** The positive relationship between Education and Income is visible, with customers at higher education levels generally appearing at higher income levels. This relationship is also reflected in the correlation analysis.
- **Usage and Miles:** The positive relationship between Usage and Miles is clearly visible. Customers who use the treadmill more frequently generally report higher expected weekly mileage, with KP781 customers concentrated toward the high-usage and high-mileage region.
- **Income and Usage:** KP781 customers are concentrated at higher income levels and higher workout frequency, while KP281 and KP481 customers show substantial overlap at lower-to-moderate income and usage levels. This indicates that income and usage together help distinguish KP781 from the other two products.
- **Income and Fitness:** KP781 customers tend to combine higher income with higher fitness levels, whereas KP281 and KP481 have greater concentration in moderate-income and moderate-fitness ranges.
- **Usage and Fitness:** Higher Usage is associated with higher Fitness levels. KP781 customers are strongly concentrated in the high-usage/high-fitness region, supporting the positioning of KP781 toward more active and fitness-oriented customers.
- **Overall product differentiation:** The pairplot shows that **KP781 is the most clearly separated customer segment**, particularly across Income, Usage, Fitness, and Miles. KP281 and KP481 have considerably more overlap, suggesting that these two products serve relatively similar customer profiles.

---

4. Missing Value & Outlier Detection

4.1 Missing Values

```
print("Missing Values:")

print(df.isnull().sum())
```

```
Missing Values:
Product          0
Age              0
Gender           0
Education        0
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64
```

Insights

- All columns contain 0 missing values.
- No missing-value treatment is required.

4.2 Outliers Using Boxplots

```
# 1. Check for missing values

print("Missing Values:")

print(df.isnull().sum())

print("\nTotal Missing Values:", df.isnull().sum().sum())


# 2. Compare mean and median to understand potential skewness

numeric_cols = ["Age", "Education", "Usage","Income", "Fitness", "Miles"]

print("\nMean vs Median:")

display(df[numeric_cols].agg(["mean", "median"]).T)
```

```python
# 3. Visualize potential outliers using boxplots
fig, axes = plt.subplots(2, 3, figsize=(15, 8))

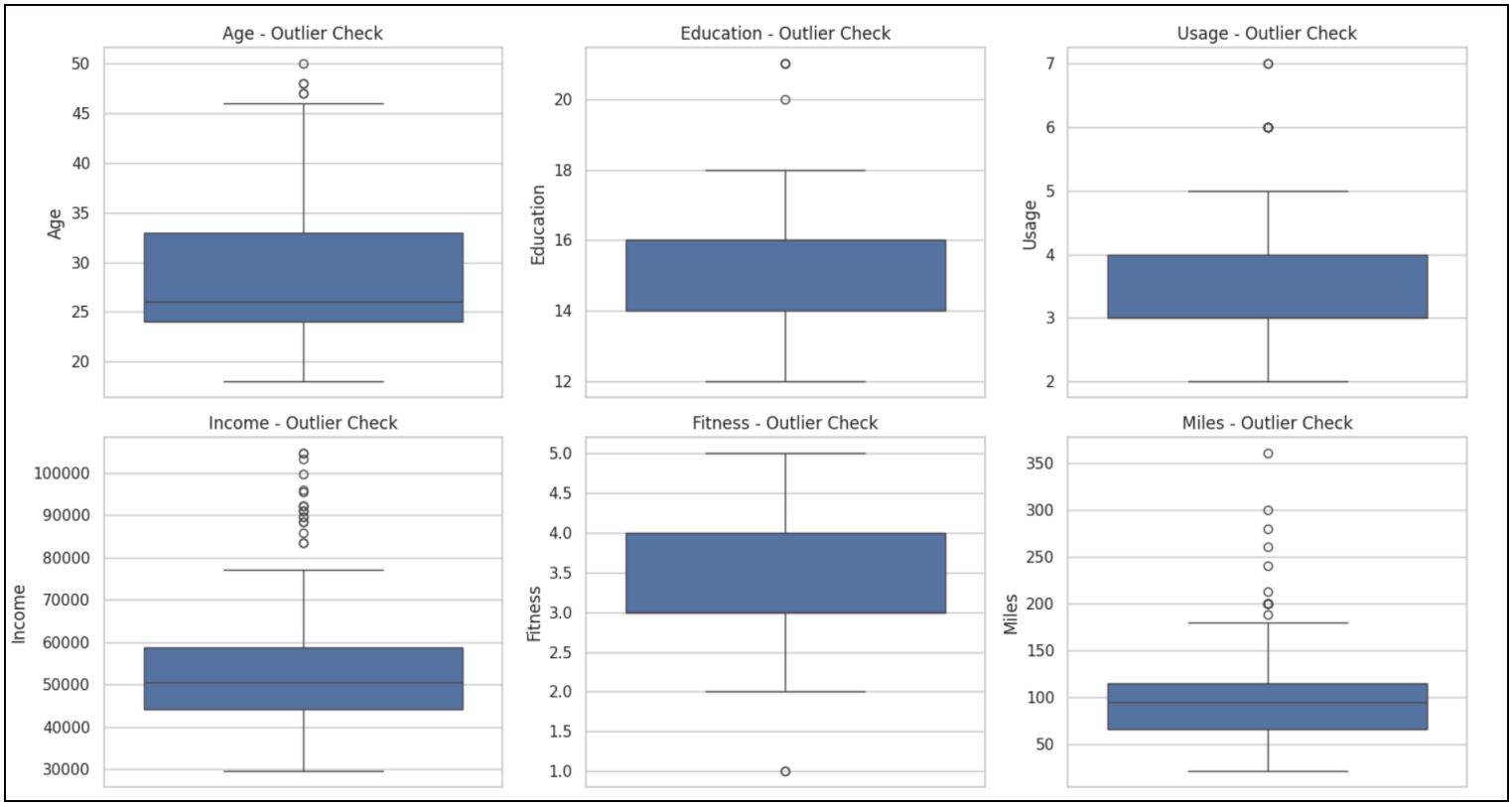
for ax, col in zip(axes.flatten(), numeric_cols):
    sns.boxplot(y=df[col], ax=ax)
    ax.set_title(f"{col} - Outlier Check")
    ax.set_ylabel(col)
plt.tight_layout()
plt.show()
```

```
Missing Values:
Product          0
Age              0
Gender           0
Education        0
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64

Total Missing Values: 0
```

Mean vs Median:

	mean	median
Age	28.788889	26.0
Education	15.572222	16.0
Usage	3.455556	3.0
Income	53719.577778	50596.5
Fitness	3.311111	3.0
Miles	103.194444	94.0



Insights

- **No missing values:** All 180 customer records are complete across the nine attributes, so no missing-value treatment is required before further analysis.
- **Income and Miles show the strongest evidence of extreme values:** Their means are higher than their medians (Income: 53,719.58 vs. 50,596.50; Miles: 103.19 vs. 94), which is consistent with the high-end observations visible in the boxplots.
- **Age also contains a small number of higher-end observations:** Most customers are younger, while a few customers are substantially older, creating potential high-end outliers.
- **Outliers should not be removed automatically:** High Income, Miles, or Age values may represent genuine customers rather than data errors. These observations should therefore be flagged and retained unless there is evidence that they are incorrect.

```
### Outlier Detection using the 1.5 × IQR Rule

outlier_summary = []

for col in numeric_cols:

    Q1 = df[col].quantile(0.25)

    Q3 = df[col].quantile(0.75)
```

```
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    outlier_count = len(outliers)
    outlier_percentage = (outlier_count / len(df)) * 100

    outlier_summary.append({
        "Attribute": col,
        "Lower Bound": lower_bound,
        "Upper Bound": upper_bound,
        "Outlier Count": outlier_count,
        "Outlier Percentage": round(outlier_percentage, 2)
    })
outlier_summary = pd.DataFrame(outlier_summary)
display(outlier_summary)
```

	Attribute	Lower Bound	Upper Bound	Outlier Count	Outlier Percentage
0	Age	10.500	46.500	5	2.78
1	Education	11.000	19.000	4	2.22
2	Usage	1.500	5.500	9	5.00
3	Income	22144.875	80581.875	19	10.56
4	Fitness	1.500	5.500	2	1.11
5	Miles	-7.125	187.875	13	7.22

Insights :

- **1.5 × IQR Rule:** Potential outliers were identified using the standard 1.5 × IQR method for the six numerical attributes.
- **Income and Miles:** These variables show the clearest extreme observations, consistent with the boxplot and mean-versus-median analysis.

- **Business Interpretation:** The extreme values in Income and Miles may represent genuine customers with higher purchasing power or higher expected treadmill usage, rather than data-entry errors.
- **Outlier Handling Decision:** The identified outliers will **not be removed** from the dataset. There is no evidence that these observations are incorrect, and removing them could eliminate meaningful customer segments.
- **Final Decision:** The original dataset will be retained for further analysis, while the identified outliers will be considered when interpreting customer profiles and product preferences.

4.3 Mean vs Median

```
mean_median = df[numeric_cols].describe().T[
    ['mean', '50%']
]
mean_median.columns = ['Mean', 'Median']
mean_median['Mean - Median'] = (
    mean_median['Mean'] - mean_median['Median']
)
display(mean_median)
```

	Mean	Median	Mean - Median
Age	28.788889	26.0	2.788889
Education	15.572222	16.0	-0.427778
Usage	3.455556	3.0	0.455556
Income	53719.577778	50596.5	3123.077778
Fitness	3.311111	3.0	0.311111
Miles	103.194444	94.0	9.194444

Insights

- **Income:** The mean income is **$53,719.58**, compared with a median of **$50,596.50**. The higher mean indicates that higher-income observations are pulling the average upward, consistent with the presence of upper-end values identified in the outlier analysis.

- **Miles:** The mean mileage is **103.19**, compared with a median of **94**. The higher mean suggests that some customers report substantially higher mileage, consistent with the upper-end outliers identified earlier.
- **Other variables:** The mean and median are relatively close for most remaining variables, indicating less influence from extreme observations compared with Income and Miles.
- **Overall:** The mean–median comparison supports the boxplot and 1.5×IQR findings that **Income and Miles contain the clearest evidence of extreme values and right-skewed distributions**. These observations are retained because they may represent genuine high-income or high-usage customers rather than data errors.

Overall Insight: The boxplots help identify potential outliers across the six attributes. These values should be investigated before taking action, as they may reflect genuine differences among customers rather than data errors.

---

5. Business Insights Based on Non-Graphical and Visual Analysis

This section brings together the findings required by the evaluation criteria, including customer profiling and probability from the case-study requirements.

Customer Profiling

```
profile = df.groupby('Product', observed=True).agg(
    Customers=('Product', 'size'),
    Avg_Age=('Age', 'mean'),
    Avg_Education=('Education', 'mean'),
    Avg_Usage=('Usage', 'mean'),
    Avg_Income=('Income', 'mean'),
    Avg_Fitness=('Fitness', 'mean'),
    Avg_Miles=('Miles', 'mean')
).round(2)
display(profile)
```

Product	Customers	Avg_Age	Avg_Education	Avg_Usage	Avg_Income	Avg_Fitness	Avg_Miles
KP281	80	28.55	15.04	3.09	46418.02	2.96	82.79
KP481	60	28.90	15.12	3.07	48973.65	2.90	87.93
KP781	40	29.10	17.32	4.78	75441.58	4.62	166.90

Insights

- **KP281 represents the largest customer segment**, with **80 customers (44.44%)**. Its average income of **$46,418**, usage of approximately **3.09 times/week**, and moderate fitness level indicate a broader customer base with moderate treadmill usage.
- **KP481 has 60 customers (33.33%)** and shows a profile broadly similar to KP281 in age, usage, and fitness. Its average income of approximately **$48,974** is only moderately higher than KP281, suggesting that KP481 does not have a sharply differentiated customer profile based on these variables.
- **KP781 has the most distinct customer profile**, with **40 customers (22.22%)** but substantially higher average income (**$75,442**), usage (**4.78 times/week**), fitness (**4.63**), and mileage (**166.90 miles**) than KP281 and KP481.
- The largest product-level differences are therefore seen in **Income, Usage, Fitness, and Miles**, rather than Age. This indicates that **customer purchasing power and activity level are stronger product differentiators than basic demographic characteristics such as age**.
- Overall, the profiles suggest three different positioning opportunities: **KP281 for the broader moderate-use customer base, KP481 for a similar middle segment, and KP781 for higher-income, frequent, and highly active customers**.

Marginal Probability

```
marginal_probability = pd.crosstab(
    df['Product'],
    columns='Percentage',
```

```
    normalize=True
).mul(100).round(2)
display(marginal_probability)
```

col_0	Percentage
Product	
KP281	44.44
KP481	33.33
KP781	22.22

Insights

- **KP281 has the highest marginal probability at 44.44%**, making it the most commonly purchased treadmill and indicating the broadest overall customer demand among the three products.
- **KP481 accounts for 33.33% of customers**, giving it a substantial share of the market and positioning it as the second most preferred product.
- **KP781 has the lowest overall customer share at 22.22%**. However, its lower marginal probability should not be interpreted as weak demand because the product serves a more specific customer profile, as shown by its higher income, usage, fitness, and mileage levels.
- The distribution shows that **KP281 and KP481 together account for 77.77% of customers**, indicating that the majority of customers purchase the two products serving the broader/moderate-use segments.
- Overall, the marginal probabilities indicate **KP281 has the broadest customer reach, while KP781 represents a smaller but more differentiated customer segment**. This distinction should be considered when planning product positioning and marketing strategies.

Conditional Probability — Male Customer Buying KP781

```
gender_product = pd.crosstab(
    df['Gender'],
    df['Product']
)
display(gender_product)
```

```
p_kp781_given_male = (
    gender_product.loc['Male', 'KP781']
    / gender_product.loc['Male'].sum()
)
p_kp781_given_female = (
    gender_product.loc['Female', 'KP781']
    / gender_product.loc['Female'].sum()
)

print(f"P(KP781 | Male) = {p_kp781_given_male:.2%}")
print(f"P(KP781 | Female) = {p_kp781_given_female:.2%}")
```

Product	KP281	KP481	KP781
Gender			
Female	40	29	7
Male	40	31	33

```
P(KP781 | Male) = 31.73%
P(KP781 | Female) = 9.21%
```

Insights

- Among **104 male customers**, **33 purchased KP781**, giving a conditional probability of **P(KP781 | Male) = 31.73%**. This means approximately **1 in 3 male customers** in the dataset purchased KP781.
- Among **76 female customers**, only **7 purchased KP781**, resulting in **P(KP781 | Female) = 9.21%**. This means fewer than **1 in 10 female customers** purchased KP781.
- The probability of purchasing KP781 is therefore approximately **3.4 times higher among male customers than female customers** (31.73% vs. 9.21%). This indicates a substantial gender difference in KP781 preference within this dataset.

- This finding is consistent with the product-level profile of KP781, where **82.50% of KP781 customers are male**. Therefore, male customers represent an important segment for KP781 marketing and targeting.
- However, gender alone should not be used to identify potential KP781 customers. The earlier analysis shows that **higher income, usage, fitness, and mileage** provide additional and stronger indicators of the KP781 customer profile.
- **Business impact:** KP781 marketing can place greater emphasis on male customers, while combining gender with **high usage, higher fitness, and higher purchasing power** to identify customers who more closely match the observed KP781 profile.

Two-Way Contingency Tables and Conditional & Marginal Probabilities

To understand how customer characteristics are associated with treadmill product choice, two-way contingency tables are constructed for **Gender vs Product** and **Marital Status vs Product**. Both conditional probabilities are calculated to understand product preference within each customer group and customer composition within each product.

```
# Gender vs Product: Two-Way Contingency Table


gender_product = pd.crosstab(df['Gender'], df['Product'])


print("Gender vs Product - Customer Counts")
display(gender_product)


# P(Product | Gender)
product_given_gender = pd.crosstab(
    df['Gender'],
    df['Product'],
    normalize='index'
).mul(100).round(2)


print("P(Product | Gender) - Percentage")
```

```
display(product_given_gender)

# P(Gender | Product)
gender_given_product = pd.crosstab(
    df['Gender'],
    df['Product'],
    normalize='columns'
).mul(100).round(2)

print("P(Gender | Product) - Percentage")
display(gender_given_product)


# Marital Status vs Product: Two-Way Contingency Table

marital_product = pd.crosstab(df['MaritalStatus'], df['Product'])

print("Marital Status vs Product - Customer Counts")
display(marital_product)

# P(Product | Marital Status)
product_given_marital = pd.crosstab(
    df['MaritalStatus'],
    df['Product'],
    normalize='index'
).mul(100).round(2)

print("P(Product | Marital Status) - Percentage")
```

```
display(product_given_marital)

# P(Marital Status | Product)
marital_given_product = pd.crosstab(
    df['MaritalStatus'],
    df['Product'],
    normalize='columns'
).mul(100).round(2)

print("P(Marital Status | Product) - Percentage")
display(marital_given_product)

# Marginal probabilities

marginal_probabilities = pd.DataFrame({
    'Probability (%)': [
        df['Product'].value_counts(normalize=True)['KP281'] * 100,
        df['Product'].value_counts(normalize=True)['KP481'] * 100,
        df['Product'].value_counts(normalize=True)['KP781'] * 100,
        df['Gender'].value_counts(normalize=True)['Male'] * 100,
        df['Gender'].value_counts(normalize=True)['Female'] * 100,
        df['MaritalStatus'].value_counts(normalize=True)['Partnered'] * 100,
        df['MaritalStatus'].value_counts(normalize=True)['Single'] * 100
    ]
}, index=['P(KP281)','P(KP481)','P(KP781)','P(Male)','P(Female)',
    'P(Partnered)','P(Single)']).round(2)

print("Marginal Probabilities")
```

```
display(marginal_probabilities)
```

Gender vs Product - Customer Counts

Gender	KP281	KP481	KP781
Female	40	29	7
Male	40	31	33

P(Product | Gender) - Percentage

Gender	KP281	KP481	KP781
Female	52.63	38.16	9.21
Male	38.46	29.81	31.73

P(Gender | Product) - Percentage

Gender	KP281	KP481	KP781
Female	50.0	48.33	17.5
Male	50.0	51.67	82.5

Marital Status vs Product - Customer Counts

MaritalStatus	KP281	KP481	KP781
Partnered	48	36	23
Single	32	24	17

P(Product | Marital Status) - Percentage

MaritalStatus	KP281	KP481	KP781
Partnered	44.86	33.64	21.50
Single	43.84	32.88	23.29

P(Marital Status | Product) - Percentage

MaritalStatus	KP281	KP481	KP781
Partnered	60.0	60.0	57.5
Single	40.0	40.0	42.5

Marginal Probabilities

	Probability (%)
P(KP281)	44.44
P(KP481)	33.33
P(KP781)	22.22
P(Male)	57.78
P(Female)	42.22
P(Partnered)	59.44
P(Single)	40.56

Insights:

- Among male customers, **38.46% purchased KP281, 29.81% purchased KP481, and 31.73% purchased KP781**. Among female customers, KP281 was the most preferred product at **52.63%**, while only **9.21% purchased KP781**.
- KP781 has a strong male customer concentration: **82.50% of KP781 customers are male**, compared with 50.00% for KP281 and 51.67% for KP481.
- Marital status shows little difference in product preference. Among partnered customers, **44.86% purchased KP281, 33.64% KP481, and 21.50% KP781**. Among single customers, the corresponding proportions were **43.84%, 32.88%, and 23.29%**.
- Within each product, the partnered share is also similar: **60.00% for KP281, 60.00% for KP481, and 57.50% for KP781**.
- The marginal probabilities show that **KP281 has the highest overall customer share (44.44%)**, followed by KP481 (33.33%) and KP781 (22.22%).
- Overall, **gender is more useful than marital status for differentiating product preference**, particularly for identifying the KP781 customer segment.

Overall Business Insights

- **Age and marital status are weak product differentiators**, as the product groups have very similar average ages and similar partnered/single proportions.
- **Income, Usage, Fitness, and Miles show stronger product-level differences**, particularly for KP781, which consistently appears at higher levels across these variables.
- **KP781 represents a distinct high-value and high-activity customer segment**, with higher average income, usage, fitness, and mileage than KP281 and KP481.
- **Usage and Miles have the strongest positive correlation at 0.76**, while Usage and Fitness also show a strong positive relationship at **0.67**. This indicates that more frequent treadmill users generally tend to report higher mileage and fitness levels.
- Overall, **activity level and income provide stronger signals for differentiating product segments than age or marital status**.

Comments on the Range of Attributes

- **Age** ranges from **18 to 50 years**, with customers concentrated mainly in their mid-20s to early 30s.
- **Income** ranges from approximately
- 29,562to
- **104,581**, showing a wide variation in customer purchasing power.
- **Usage** ranges from **2 to 7 times per week**, with most customers concentrated around **2–4 times per week**.
- **Fitness** ranges from **1 to 5**, with **Fitness level 3** being the most common.

- **Miles** ranges from **21 to 360 miles**, with most customers at lower-to-moderate mileage and a smaller group reporting substantially higher mileage.

Comments on the Distribution of Variables and Relationships

- **Income and Miles are right-skewed**, with higher-end observations contributing to the longer right tails. This is consistent with the outlier analysis performed earlier.
- **Usage and Miles have the strongest positive relationship (0.76)**, indicating that customers who use the treadmill more frequently generally report higher mileage.
- **Usage and Fitness are positively correlated (0.67)**, showing that higher-frequency users tend to have higher fitness ratings.
- **Education and Income are also positively related (0.63)**, indicating that higher education levels are associated with higher income in this dataset.
- The relatively weak relationship between **Age and Miles (0.03)** supports the earlier finding that age is not a strong differentiator of treadmill product choice.

Comments on Univariate and Bivariate Plots

- **Product distribution:** KP281 is the most purchased product, accounting for **44.44%** of customers, followed by KP481 at **33.33%** and KP781 at **22.22%**.
- **Gender and Marital Status:** The customer base is slightly male-dominated (**57.78%**) and more customers are partnered (**59.44%**), but these characteristics alone do not strongly differentiate KP281 and KP481.
- **Continuous-variable distributions:** Usage is concentrated around **2–4 times per week**, Fitness is centered around level **3**, while Income and Miles show right-skewed distributions with higher-end observations.
- **Marital Status vs Product:** Partnered customers account for approximately **57.5%–60% across all products**, indicating little product preference difference based on marital status.
- **Age vs Product:** Product groups have very similar age profiles, with average ages between **28.55 and 29.10 years**, indicating that age is not a strong product-selection factor.
- **Income, Usage, Fitness, and Miles vs Product:** KP781 consistently occupies the higher end of these measures, creating the clearest product-level segmentation in the analysis.
- **Correlation Heatmap:** Usage–Miles (**0.76**) and Usage–Fitness (**0.67**) are the strongest relationships, highlighting customer activity as an important underlying pattern.
- **Pairplot:** KP781 forms the most distinct cluster across **Income, Usage, Fitness, and Miles**, while KP281 and KP481 show greater overlap. This supports the conclusion that KP781 serves a more specialized, higher-income and higher-activity customer segment.

---

6. Recommendations

- **Position KP281 as the broad-market product**, as it has the largest customer share at **44.44%** and appeals to a broader, moderate-use customer base.
- **Target KP781 toward higher-income customers**, as its average customer income is **$75,442**, substantially higher than KP281 and KP481.
- **Focus KP781 marketing on frequent and highly active users**, as KP781 customers average **4.78 workouts per week**, a **4.63 fitness score**, and **166.90 miles**.
- **Use usage, fitness, and mileage together for KP781 targeting**, as these characteristics provide a stronger distinction between KP781 and the other products than age or marital status.
- **Consider male customers as an important KP781 segment**, since **82.50% of KP781 customers are male**. However, gender should be used as a supporting targeting factor rather than the only selection criterion.
- **Do not use age as the primary product-targeting factor**, because average age is very similar across products, ranging from **28.55 to 29.10 years**.
- **Do not use marital status as a primary targeting factor**, because the partnered share is similar across products at approximately **57.50%–60.00%**.
- **Position KP481 as a middle-segment option**, as its customer profile is broadly similar to KP281 but represents a smaller **33.33%** of the customer base.
- **Use customer activity and purchasing profile to differentiate marketing messages**, emphasizing accessibility and moderate usage for KP281 while highlighting performance and higher activity levels for KP781.
- **Maintain distinct product positioning:** KP281 for the broader moderate-use market, KP481 for the middle segment, and KP781 for higher-income, frequent, and highly active customers.